

Jan de Priester

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PROFESSIONAL SUMMARY

Ph.D. Candidate & Safe RL Researcher with 5+ years of experience designing learning-based control architectures (Python/PyTorch) backed by verifiable safety and robustness guarantees. Specializes in guaranteeing the **safety and reliable deployment of RL-agents** against adversarial edge cases. First-author researcher (**RLC** 3x, **ACC**) and RLC Senior Reviewer targeting a Winter 2027 Research Scientist internship in the SF Bay Area.

TECHNICAL SKILLS

Core Competencies: Machine Learning (ML), Reinforcement Learning (RL), Continuous/Hybrid Control Theory, Stochastic Optimization, Simulation

Programming Languages: Python (PyTorch, JAX), Git, Docker, MATLAB, L^AT_EX

EXPERIENCE

Machine Learning Researcher (Ph.D.) Sep. 2022 – Present

Hybrid Systems Laboratory, University of California, Santa Cruz Santa Cruz, CA

- Engineered an RL algorithmic framework formally framing the RL problem to mathematically **guarantee system stability** without imposing constraints on the underlying learning process (*Published: **RLC***).
- Constructed AI supervisory hysteresis mechanisms using **high-dimensional latent features**, achieving guaranteed robustness to adversarial perturbations in edge cases (*Target: **IEEE T-RL***).
- Formulated a predictive hybrid logic supervisor to formally **guarantee the robust orchestration** between lower-level policies and skills, enabling **reliable deployment** by eliminating edge case failures while ensuring global stability (*Published: **RLC***).
- Mentored and directed** a Master's researcher through an experimental robotics validation project, directly formulating the theoretical hysteresis logic and mathematical bounds while guiding technical writing for physical hardware deployment (*Published: **IEEE CCTA***).

EDUCATION

University of California, Santa Cruz Santa Cruz, CA

Ph.D. in Electrical and Computer Engineering (Advisor: Prof. Ricardo G. Sanfelice) Sep. 2022 – Expected Aug. 2027

Eindhoven University of Technology Eindhoven, Netherlands

M.Sc. in Mechanical Engineering Sep. 2018 – Nov. 2020

Eindhoven University of Technology Eindhoven, Netherlands

B.Sc. in Mechanical Engineering Sep. 2015 – Jul. 2018

SELECTED PUBLICATIONS

- Jan de Priester** and Ricardo G. Sanfelice. "Hysteresis-Based Reinforcement Learning from Latent Representations for Robust Obstacle Avoidance". *In Preparation, Target IEEE Transactions on Reinforcement Learning (**T-RL**)* (2026).
- Jan de Priester** and Ricardo G. Sanfelice. "Intrinsic Closed-Loop Practical Asymptotic Stability in Standard Reinforcement Learning". *Reinforcement Learning Journal (**RLC/RLJ**) TBD* (2026), pp. TBD.
- India Spott, **Jan de Priester**, and Ricardo G. Sanfelice. "An Experimentally Validated Hybrid Control Strategy for Robust Contact Detection and Force Regulation". *IEEE Conf. on Control Tech. & Applications (**CCTA**)*. 2026.
- Jan de Priester** and Ricardo G. Sanfelice. "A Timer-Based Hybrid Supervisor for Robust, Chatter-Free Policy Switching". *Reinforcement Learning Journal (**RLC/RLJ**) 6* (2025), pp. 2508-2529.
- Jan de Priester**, Zachary Bell, Prashant Ganesh, and Ricardo G. Sanfelice. "MultiHyRL: Robust Hybrid RL for Obstacle Avoidance against Adversarial Attacks". *Reinforcement Learning Journal (**RLC/RLJ**) 4* (2024), pp. 2017-2040.
- Jan de Priester**, Ricardo G. Sanfelice, and Nathan Van De Wouw. "Hysteresis-Based RL: Robustifying Reinforcement Learning-based Control Policies". *IEEE American Control Conference (**ACC**)*. 2022.

HONORS, SERVICE & PRESENTATIONS

Leadership | Holistic Reviewer (Senior Reviewer), **RLC** (2026); Technical Reviewer, **RLC** (2025), ACM **HSCC** (2025), **NAHS** (2023)

Talks & Presentations | Northern California Control Workshop (**NorCal**) (2026); Reinforcement Learning Conference (**RLC**) (2025, 2024); Bay Area Robotics Symposium (**BARS**) (2022)

Awards | Chancellor's Fellowship, UCSC (2022); **1st Place**, Business Process Intelligence Challenge (2019); **1st Place**, Shell Energy Challenge (2018)